



End Term (Even) Semester Regular/Back Examination May 2026

Roll no. 2294038

Name of the Program and semester: **B. Tech.(CSE)-VIIIth Semester**

Name of the Course: **Disaster Management**

Course Code: **TDM-881**

Time: 3 hours

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

Q1.

(2X10=20 Marks)

- a. Explain the man-induced disasters such as water logging, land subsidence, groundwater depletion, soil erosion, release of toxic gases, hazardous chemicals, and nuclear explosions. Also explain their causes, impacts, and preventive strategies. (CO2)
- b. Explain the phenomenon of cloud bursts, landslides, and avalanches, particularly in Himalayan regions. Discuss their causes, effects, and also define mitigation strategies. (CO2)
- c. Analyze the classification of disasters into natural and man-induced categories. Provide detailed examples and explain their distinguishing features. (CO3)

Q2.

(2X10=20 Marks)

- a. Design without sustainability increases disaster risks." Critically examine this statement with suitable examples and suggest measures for balanced development. (CO4)
- b. Explain the functions and responsibilities of Panchayati Raj Institutions (PRIs) and Urban Local Bodies (ULBs) in disaster management. (CO5)
- c. Explain the inter-relationship between disasters and development. How can development both increase and reduce disaster risks? (CO4)

Q3.

(2X10=20 Marks)

- a. Explain how Information, Education, and Communication (IEC) strategies contribute to disaster preparedness and awareness. (CO5)
- b. Discuss the role of hazard zonation mapping in disaster preparedness. How does it help in risk reduction and planning? (CO6)
- c. Analyze the need for population reduction or relocation in vulnerable areas. What are the challenges involved? (CO3)



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Q4.

(2X10=20 Marks)

- a. Analyze the challenges faced during the emergency response phase and suggest solutions with the help of suitable example as per given challenges. (CO3)
- b. "Strong policies and institutions are the backbone of effective disaster management. "Critically examine this statement with reference to India. (CO6)
- c. Describe the structure, composition, and functions of the National Disaster Management Authority (NDMA). (CO4)

Q5.

(2X10=20 Marks)

- a. Explain the role and organizational structure of the National Disaster Response Force (NDRF). (CO4)
- b. Describe the functions and contributions of the National Institute of Disaster Management (NIDM) in capacity building and research. (CO4)
- c. Explain the following term: -
 - (i) Disaster Management Act, 2005 (CO5)
 - (ii) Nuclear Explosions. (CO1)
 - (iii) Droughts Cyclones (CO1)
 - (iv) Diversion of river routes (ex. Kosi river) in natural disaster (CO1)